**11** The height $h$ cm of a sunflower plant $t$ days after planting the seed is modelled by $h = a + b \ln t$ for $t \geqslant 9$, where $a$ and $b$ are constants. The sunflower is 10 cm tall 10 days after planting and 200 cm tall 85 days after planting.

**(a) (i)** Show that the value of $b$ which best models these values is 88.8 correct to **3** significant figures. **[2]**

**(ii)** Find the corresponding value of $a$. **[1]**

**(b) (i)** Explain why the model is not suitable for small positive values of $t$. **[1]**

**(ii)** Explain why the model is not suitable for very large positive values of $t$. **[1]**

**(c)** Show that the model indicates that the sunflower grows to 1 m in height in less than half the time it takes to grow to 2 m. **[2]**

**(d)** Find the value of $t$ for which the rate of growth is 3 cm per day. **[3]**

| | |
|---|---|
| **11(a)(i)** | |
| **11(a)(ii)** | |
| **11(b)(i)** | |
| **11(b)(ii)** | |

**11(c)**

**11(d)**